Beyond Dark Patterns in Games: A Radiant Framework for Sustainable Player Well-being

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Abstract. Digital games support well-being, yet dark patterns often frustrate player needs for autonomy, competence, and relatedness. Radiant Patterns (RPs) offer an ethical alternative but remain theoretical. This paper applies RPs using workshops and structured analysis of commercial games. We provide three practical tools: a refined lexicon, a tiered model of levels and variants, and a documentation template. We also propose a framework for a collaborative catalog with a governance model to enable future community growth. This infrastructure helps bridge the research-industry gap by balancing player well-being with commercial viability.

Keywords: Radiant Patterns · Ethical Game Design · Player Well-being · Self-Determination Theory · Dark Pattern Countermeasures

1 Introduction

Digital games have a dual nature. They can provide engaging and uplifting experiences [13,11,16], offering relaxation, therapeutic outcomes, and social benefits [29,39]. Yet, this potential is compromised when games are misused, including excessive play linked to behavioral issues [33], toxicity in online communities [52], and exploitative micro-transactions [32]. Such practices undermine well-being by frustrating basic psychological needs [18] through dark or deceptive design patterns [26,53], leading to negative emotional states, addictive behaviors [7], and gaming disorder [31].

Dark Patterns (DPs)[1] are interface techniques that obscure agency and prioritize provider utility over autonomy [5]. Regardless of intent, they undermine informed decision-making [25], nudging users toward data surrender, unwanted retention, or unsolicited purchases by exploiting heuristic vulnerabilities [5,35]. In games, DPs extend beyond isolated dialogs, embedding into progression, reward, and monetization systems [56], often through aggressive practices [44].

[1] We distinguish between Dark and Deceptive: the latter implies misleading techniques, while the former includes manipulative and coercive patterns. We use Dark Patterns to encompass the full spectrum of designs that limit user agency.

To mitigate these effects, HCI research has proposed countermeasures [17,50]. Bright Patterns [49] and Fair Patterns [19] address general interfaces, while Positive Computing promotes technologies that support psychological well-being [9]. In games, Radiant Patterns (RPs) were introduced as an ethical alternative to DPs [38]. Grounded in Self-Determination Theory (SDT) [18], RPs define game mechanics that support player well-being within design practice.

Despite this foundation, RPs face a deployment gap. Although conceptually defined [38] and tested in isolation [20], they remain largely academic. Unlike DPs, which are documented to drive retention and profit, RPs have lacked the structural guidance needed to align with commercial production constraints. Designers have theory but no shared resource for refinement. Building a living repository is necessary. This effort connects research and industry through shared tools for healthier gaming. It meets the Ethical Games movement's call [27] for evidence-based standards. While regulation protects players, it lacks technical guidance for developers. The Radiant Patterns establishes the infrastructure and governance to turn ethical principles into design standards for player wellbeing.

This work presents a refined Radiant Pattern framework and a collaborative public catalog. We describe the methodology used to expand the initial set and introduce an online platform designed as scalable infrastructure for ongoing contributions. The catalog provides an initial corpus of RPs, a standardized documentation template, and a governance model to support submission and review. To achieve this, we followed a three-step process: (I) collaborative workshops to critically evaluate the initial set and identify new potential RPs based on diverse gaming experiences; (II) curated refinement sessions aimed at consolidating definitions and ensuring rigorous alignment with the RP framework; and (III) an exploratory autoethnographic study where the research team systematically analyzed popular games using the RP framework to document instances of positive design and categorize new candidate patterns.

The main contribution of this work is transforming RPs into a living, scalable resource that provides practitioners with the professional vocabulary to advocate for well-being-supportive mechanics over profit-driven deceptive defaults. By establishing a collaborative methodology, an initial set of validated Radiant Patterns, and a public platform, we empower researchers and practitioners to move beyond merely identifying DPs to actively participating in building a repository of positive, well-being-supportive alternatives, thereby promoting a design culture grounded in respect for the player.

2 Background

Research in digital games has used design patterns to identify recurring structures in gameplay systems [4]. Patterns serve as conceptual abstractions connecting mechanics, player interactions, and experiential outcomes, enabling designers to record and share knowledge systematically [48,4]. However, as games have become embedded in commercial, social, and psychological contexts, pattern analysis has expanded beyond functional concerns. Patterns are not neutral

solutions—they shape player behavior, influence decisions, and mediate player-system relationships [56,38]. This has motivated critical research examining how pattern-based approaches obscure user choices [56,22], exploit cognitive vulnerabilities [15,40], and limit agency in apps and games [25,22]. Concurrently, constructive research has proposed design approaches prioritizing ethical interaction [8,38,43,27]. This background addresses these complementary perspectives.

2.1 Well-being and "Positive" Design Patterns

In contrast to manipulative strategies, *positive computing* positions well-being as an explicit design objective rather than a byproduct of usability [8,9,42]. It draws on multiple strands of psychological research to inform the design of systems that support human flourishing [9,43]. Among these, Self-Determination Theory (SDT) provides a widely adopted and empirically robust framework for understanding basic psychological needs. SDT identifies autonomy, competence, and relatedness as fundamental needs [18,47], linking need satisfaction to intrinsic motivation and growth, and need frustration to diminished well-being [47]. In design terms, these needs function as structural affordances: systems cannot guarantee flourishing, but they can organize conditions that support or undermine volition, efficacy, and connection [43].

Although Positive Computing and SDT provide a theoretical basis, restorative design vocabularies remain fragmented. We compare three approaches – Bright, Fair, and Radiant patterns – to clarify their scope and goals.

Bright patterns use persuasive design to support users' genuine goals rather than provider profit motives [49]. They reframe ambiguous persuasive techniques to prioritize user welfare, for example through transparent onboarding or visual countermeasures against scarcity cues. However, this approach introduces a normative tension between "nudging for good" and respecting autonomy, as steering users toward desirable behaviors may risk ethical paternalism.

Fair patterns emphasize neutrality and regulatory compliance rather than persuasion. They are defined as design solutions that restore fairness by providing clear and timely information to facilitate informed decision-making [45]. Their goal is to operationalize transparency and trust through mechanisms such as layered notices and non-coercive defaults, with primary application in regulated domains such as e-commerce and digital services.

Radiant patterns (RPs) are explicitly tailored to game design and grounded in SDT. They aim to discourage addictive play and promote well-being by supporting SDT basic psychological needs [21]. Unlike the broader applicability of Bright or Fair patterns, RPs address specific gameplay structures. For example, autonomy support involves making meaningful choices visible within the system, while competence scaffolds foster skill-based mastery rather than artificial retention loops [20]. This domain specificity allows RPs to target the intrinsic quality of the experience, not only the transparency of the interface.

Because behavioral design inevitably shapes user action regardless of intent [3], ethical concerns extend beyond whether persuasion is used and concern how

systems organize motivation. Many positive design initiatives risk focusing narrowly on individual symptoms of harm while neglecting broader systemic causes [46]. RPs address this by emphasizing the structural non-frustration of basic psychological needs, establishing a pragmatic ethical baseline for game design rather than prescribing idealized outcomes.

A further limitation across positive pattern approaches is the scarcity of empirical validation compared to the iterative A/B testing commonly used to optimize dark patterns. Without standardized definitions and shared criteria, consistent implementation and evaluation remain difficult. Standardizing definitions and identification criteria provides the experimental units required for large-scale testing. This framework establishes the consistent variables needed for cumulative empirical research to assess impact on player experience.

2.2 Radiant Patterns

Radiant Patterns were introduced as an ethical, pattern-based approach to game design in response to the prevalence of dark patterns and their negative impact on player experience [38]. Rather than only cataloging harmful practices, this work articulated recurring design solutions to guide developers toward ethically grounded alternatives compatible with production constraints such as progression systems and monetization models. Initially, RPs were broadly defined as patterns intended to support player well-being.

However, the emphasis on "preserving well-being" was considered too abstract to guide consistent application, particularly since meaningful play often involves tension and challenge [20]. The definition was therefore reframed to focus on the non-frustration of psychological needs. Radiant Patterns are interaction patterns intentionally implemented to ensure foundational player well-being is not significantly undermined. Specifically, RPs avoid decreasing autonomy, relatedness, and competence. They are conscious design choices that minimize predictable psychological harm and avoid patterns known to frustrate these needs, establishing ethical boundaries for player experience.

This shift from idealized flourishing to minimizing predictable harm demonstrates how Radiant Patterns can coexist with commercial production goals. A standardized template supports this by requiring explicit descriptions of how each pattern avoids need frustration [20].

An initial set of six RPs previously outlined served as exemplars for this approach [38,20]. While these patterns have been applied in reflective design artifacts [21], the methodology for expanding this set remains underdeveloped. The existing documentation template supports the recording of known patterns but offers little guidance on the identification or derivation of new ones. Consequently, the extension of the RP library has relied heavily on ad-hoc authorial judgment, limiting its transferability to the broader research community [20].

This methodological gap compounds the empirical limitation identified in the previous section. Without a structured process to generate and standardize new patterns, the lexicon of Radiant Patterns remains too small and inconsistent to support rigorous A/B testing against dark patterns. Therefore, the primary

contribution of the present work is to move beyond isolated exemplars by establishing a collaborative infrastructure and a structured methodology for the identification, refinement, and validation of Games Radiant Patterns.

3 Methodology

To develop a refined and collaborative catalog of Game Radiant Patterns, we conducted a three-phase qualitative process: (I) a Collaborative Workshop, (II) Curated Refinement Sessions, and (III) an Exploratory Autoethnography (Figure 1). Across phases, we collected qualitative materials including workshop records, session notes, and researcher diaries and memos.



Fig. 1. Overview of the methodology to expand and refine the Radiant Pattern catalog.

Nine participants took part across the three phases, all with academic and/or professional experience in digital games (Table 1). Phase I involved five participants (P01–P05). Phases II and III included five participants (P01, P06–P09). After Phase I, P01 joined the refinement and analysis stages due to prior involvement with the RP framework. Participant selection prioritized expertise-led stress testing over broad-scale validation to ensure the framework's internal logic was technically sound.

3.1 Phase I: Collaborative Workshop

In Phase I, we gathered critical feedback on the initial six Radiant Patterns and generated new candidate RPs grounded in participants' gameplay experiences. To do so, we conducted a collaborative workshop using the World Café method [6], which supports structured yet open dialogue among participants with backgrounds in HCI and game design.

The workshop unfolded in two stages: context setting and roundtable discussion. We began with a presentation of the operational definition of Radiant Patterns and the documentation template [20], ensuring a shared understanding of RP structure. Participants were then divided into two tables, each assigned three patterns from the initial set [20]. One table examined Creative Suite (formerly World Creation), Progression (formerly Item Evolution), and Adaptive Challenge (formerly Different Difficulty); the other reviewed Open Gate (formerly Free Exit), Visual Support (formerly Color Hint), and Companion (formerly Free Pet Care). Each 30-minute round centered on the question: *"Is the*

Table 1. Participant profile across the three methodology phases

P#	Age	Gen	Education Level	Study Field	Relation to Digital Games	Phase
P01	21–25	M	Undergraduate student	Games and HCI	developer, designer, player	I–II–III
P02	21–25	M	Master's student	Hate Speech and Digital Games	researcher, player	I
P03	21–25	M	Master's student	HCI	developer, researcher, player	I
P04	36–40	M	Master degree	Games and HCI	developer, researcher, player	I
P05	36–40	F	Master degree	Games, HCI, and SDT	developer, designer, researcher, player	I
P06	21–25	F	Undergraduate student	Digital Media Systems and HCI	researcher, player	II–III
P07	26–30	M	Master degree	Computer Science and HCI	developer, designer, player	II–III
P08	21–25	M	Undergraduate student	Digital Media Systems and HCI	designer, student, player	II–III
P09	26–30	M	Undergraduate student	Digital Media Systems, HCI, and SDT	designer, player	II–III

structured RP consistent with the operational definition of a RP?" Afterward, participants rotated tables to review and extend prior notes[2].

We audio-recorded and transcribed discussions and collected written notes from each table. We compiled all materials into a structured spreadsheet to support systematic comparison across patterns and rounds. Using the RP documentation template as an analytical lens, we conducted a deductive reflexive thematic analysis [14]. We focused on recurring structural issues, including unclear template elements, ambiguous pattern boundaries, inconsistencies with the operational definition, and proposed alternative formulations. We grouped issues by pattern and coded them by type, examining their distribution across groups and rounds to prioritize reformulation. This process produced a consolidated set of revision points that informed the Phase II Refinement Sessions.

3.2 Phase II: Curated Refinement Sessions

In Phase II, we focused on refining the initial proposals. We used feedback from Phase I to clarify ambiguities and address implementation concerns raised during the workshop, consolidating these insights into a revised structure and updated documentation template. Each participant took primary responsibility for one pattern, except P09, who oversaw two. We distributed assignments to balance familiarity with the RP framework and disciplinary diversity: P01 (Creative Suite), P06 (Visual Support), P07 (Companion), P08 (Progression), and P09 (Open Gate and Adaptive Challenge).

To advance this refinement, we combined individual revision with collective discussion. Participants first revisited their assigned patterns in light of Phase

[2] Distribution of Radiant Patterns across discussion rounds available in our dataset

I feedback, annotating issues of clarity, example adequacy, and alignment with the operational definition. We then held one-hour sessions dedicated to one or two patterns, debating formulations and examining boundary cases.

Decisions reached by consensus established a logical baseline, though we recognize this expert-derived agreement is a starting point for wider validation. When disagreements emerged, we documented and revisited them until reaching a stable formulation. Session notes and version history recorded the process. This phase resulted in a revised set of Radiant Patterns and an updated documentation template, which we then applied in Phase III.

3.3 Phase III: Exploratory Autoethnography

In Phase III, we applied the refined framework through exploratory autoethnography [14], using researcher gameplay as data to structurally identify how Radiant Patterns are configured in practice and to identify boundary cases for catalog refinement.

We selected games based on Phase II citations and researcher familiarity, requiring a minimum of 10 hours of play or narrative completion. This duration served as a targeted sampling period to identify specific pattern instances rather than an exhaustive audit of entire game. Adopting a dual player–analyst role, we examined how specific design elements shaped experiences of autonomy, competence, and relatedness. We focused on moments of friction, confusion, or breakdowns in support mechanisms, documenting how design structures were categorized and mapped to psychological need support during play.

Each researcher was assigned one game; due to an uneven number of researchers and titles, one participant analyzed two Table 2 summarizes the games played by each participant. Each researcher analyzed their assigned game through the lens of its corresponding design pattern, with approximately 10 hours of gameplay per title.

Table 2. Games Played per Participant and Assigned Radiant Pattern

Partic.	**Game**	**Assigned Pattern**	**Playtime**
P01	*Celeste*	Open Gate	over 10 hours
P06	*God of War: Ragnarok*	Adaptive Challenge	over 10 hours
P07	*Shadow of the Tomb Raider*	Visual Support	over 10 hours
P07	*Minecraft*	Creative Suite	over 10 hours
P08	*Teamfight Tactics (TFT)*	Companion	over 10 hours
P09	*Metroid Dread*	Progression	over 10 hours

We used structured autoethnographic diaries [14] to record reflective accounts linked to specific RPs. Entries included (i) situational descriptions, (ii) interpretations of design influence, (iii) reflections on need support or frustration, and (iv) alternative design applications. Narrative accounts situated observations within broader play arcs, while analytical memos and screenshots captured vi-

sual evidence and real-time insights. A shared template organized the materials to ensure coherence while acknowledging researcher analytical perspective.

We analyzed the data through a reflexive thematic approach grounded in the RP definition, identifying recurring patterns, ambiguities, and design elements satisfying Phase II invariants that aligned with or diverged from intended need support. This phase generated illustrative cases and clarification notes that informed further refinement and stabilization of the Radiant Pattern catalog.

4 Findings and Decurrent Improvements

This section presents results from the collaborative workshop, curated refinement sessions, and exploratory autoethnography. Game titles appear in italics (e.g., *Super Mario Maker*), and pattern names are underlined (e.g., Creative Suite).

4.1 Baseline Radiant Patterns Required Structural Refinement

Workshop results revealed that while original patterns had the necessary information, the template organization obscured alignment with RP definitions. Participants emphasized the need to revise titles for specificity, structure patterns into levels for genre flexibility, and improve the template to clarify components.

Two participants (P01, P02) highlighted the need to revise the titles of the initial set for quick and clear identification of what each pattern addressed, particularly for Free Pet Care, Different Difficulty, and World Creation to support quicker identification. As one noted: '[P02]:*we were revising it to make it more related to the idea of a companion rather than a pet, because "pet care" strongly evokes the idea of taking care of it.*'

Descriptions also lacked explicit references to basic psychological needs, hindering practical use. This concern is reflected in P01's statement: '[P01]:*It is important for the title to be more specific. It should be more distinctive and refer to the names of the elements from Self-Determination Theory*'. These issues required changes across the entire set to ensure consistency and standardization.

Although examples were generally clear, four of five participants argued that structuring patterns into levels was necessary due to cross-genre complexity. P04 noted difficulty applying Different Difficulty to souls-like games[3]: "[P04]: *Dark Souls style games are meant to be difficult. So it's not only about the developer's goal of profit or engagement, you also have to consider the narrative objectives that the game itself proposes.*" P02 similarly argued that World Creation was too broad: "[P02] *World Creation involves many things. For example, if I say 'allow players to create scenarios using a set of tools', that's one thing. Customization starting from something that already exists, is another.... There are particularities within these interactions that also address different elements.*"

To accommodate these contexts, participants proposed a tiered structure inspired by WCAG [55]: '[P01]: *Like in WCAG, where you have Level A, AA*

[3] The term "Soulslike" describes games that resemble the Dark Souls series, typically featuring clear goals combined with high difficulty and multiple obstacles.

and AAA. Level AA includes Level A and adds another element, with Level A being the basic level".

Finally, all participants identified textual organization issues, necessitating an **updated radiant pattern template**. The original structure obscured application logic: "[P01]: *I need to be able to read and clearly understand what I have to do, what I need to develop... what I, as a developer, need to do to make this pattern work.*" Participants also identified missing development goals: "[P02]: *One thing I felt was missing is which development goals it contributes to. What development objective does this pattern help achieve?*" and vague SDT links: "[P04]: *It is necessary to explain how the development goal relates to Self-Determination Theory in a concise and compact way.*"

These findings necessitated reformulating pattern presentation and Radiant template scope to improve clarity and provide a framework that explicitly addresses developer needs while protecting player well-being.

4.2 Refined RPs Framework Established a Shared Lexicon

Through a deductive thematic analysis of the workshop transcripts cross-referenced with the refinement sessions we identified recurring criteria that let to a restructured RP framework, including revised titles, new levels, and template adjustments.

Establishing How to Name a Radiant Pattern Discussion sessions revealed that original titles were often too narrow or lacked alignment with the functional depth of the patterns. Analysis of the lexical shifts produced four pragmatic naming rules to guide future expansion:

1. **Functional Abstraction**: Use terms that describe the interaction state rather than specific menus, labels, or literal UI elements or actions.
2. **Generic Classification**: Replace specific elements with broader functional categories to ensure cross-platform neutrality.
3. **Outcome Orientation**: Focus the title on the systemic or psychological result of the interaction rather than the specific object or mechanic involved.
4. **Relational Characterization**: Define the system's active response to the player's status or skill rather than using static comparative descriptors.

These rules were applied to connect the titles of the baseline set of RPs directly to their functional descriptions while providing a scalable protocol for future repository expansion. The resultant changes were:

RP1 - **Free Exit** changes to **Open Gate**: The original title emphasized freedom to leave but did not convey progress preservation. The revised name highlights both autonomy and progress security, adopting Open Gate as a metaphor for seamless exit and return without cost.

RP2 - **Free Pet Care** changes to **Companion**: The term *pet* was too narrow to cover the variety of companions in games. Companion broadens the scope to include animal, human, technological, or symbolic entities with decorative, narrative, or functional roles.
RP3 - **Item Evolution** changes to **Progression**: The original name restricted the pattern to items. It was reframed to emphasize perceived progress and achievement, decoupling it from exclusive item-based mechanics.
RP4 - **World Creation** changes to **Creative Suite**: The initial title centered on world-building and did not reflect the range of creative experiences in games. Creative Suite broadly encompasses player-generated content such as customization, level design, system creation, and worldbuilding.
RP5 - **Color Hint** changes to **Visual Support**: Hint risked being interpreted as tutorial guidance rather than gameplay affordances. Visual Support broadens the scope beyond color cues and better reflects its role in mitigating visually blocking elements.
RP6 - **Different Difficulty** changes to **Adaptive Challenge**: The original name suggested only player-controlled difficulty adjustments. The revised title captures both manual adjustments and adaptive responses to different player profiles and play styles.

By introducing these rules and applying them to the baseline RP set, we promote clearer pattern identification without restricting patterns to specific contexts, enabling expansion across settings. This supports future contextual variants while preventing ambiguity that could hinder practical application.

Defining Hierarchical Scaling and Contextual Variants for RPs Discussion revealed that original RPs failed to account for genre-specific constraints, platform limitations, or player profiles. Contextual differences, such as the transition from single-player to multiplayer, are not isolated exceptions but structural adaptations required to maintain the pattern's efficacy. By analyzing these requirements, we established two pragmatic rules for framework expansion:

1. **Progressive Requirement Scaling**: Structure pattern application into hierarchical tiers, each representing increasing psychological need satisfaction from a baseline of non-frustration.
2. **Environment-Based Subclassing**: Define structural variants that adapt universal levels to specific gameplay contexts (e.g., genre, mode, platform) without changing core logic.

These rules allow a more precise integration across varied RP titles. For instance, the core mechanic of Companion requires different implementations for decorative mascots in *The Sims* compared to combat partners in *World of Warcraft*. Similarly, Creative Suite spans a broad design space—from aesthetic customization to complex scenario sharing—that cannot be addressed by a single description without losing developer utility.

Following progressive scaling, patterns are organized into universal application levels. Level L1 establishes the ethical floor of non-frustration. While L1 represents standard practice, i.e., its inclusion in the framework identifies it as a necessary boundary against deceptive monetization or retention tactics, L2 and L3 progressively enhance need satisfaction. For example, Open Gate includes L1 – Safety (automatic progress saving), L2 – Autonomy (intentional stopping), and L3 – Competence (visual feedback reinforcing achievement).

The use of environment-based subclassing through variants ensures extensibility. This dual structure allows practitioners to add contextual implementations to the RP catalog while preserving each pattern's conceptual identity.

An Updated Radiant Pattern Framework Workshop evaluations and refinement sessions confirmed that original pattern descriptions lacked the structural rigor required for industry application. The primary failure involved the conflation of developer goals with player outcomes, which obscured how specific design choices support ethical objectives. Furthermore, the absence of explicit dimensions like genre or platform prevented the framework from capturing the structural adaptations necessary for diverse play environments. Table 3 presents the resulting template. This systematic structure reorganizes the Radiant Pattern Framework, ensuring that every RP added to the catalog supports consistent application and future extensibility without fragmenting conceptual integrity.

To ensure scalability of the RP framework, we defined four pragmatic rules for the RP template:

1. **Goal–Outcome Decoupling**: Separate developers' technical goals from **Player Problems** (the experiential goals or need frustrations addressed).
2. **Multi-Dimensional Scoping**: Specify target actors and context to avoid vague terminology and support diverse settings.
3. **Invariant–Variant Duality**: Define core properties (Invariants) that must hold across adaptations to preserve identity, regardless of the Variant.
4. **Traceable Need Satisfaction**: Map each solution to hierarchical design levels (L1–L3) to validate radiance through structural criteria.

4.3 Eliciting RPs from Play Experiences with Commercial Games

Autoethnographic results[4] confirm that these patterns already appear within successful commercial titles, though their presence does not yet validate the framework as a production tool. *Celeste* [10] yielded eight instances: three for Open Gate, four for Adaptive Challenge, and one for Companion. A researcher diary entry noted: '*the dialogues contribute a lot to building the relationship*'. While the current iteration prioritizes levels for autonomy and competence, these dialogue-driven instances confirm the relatedness dimension as a priority for future lexicon expansion. While this iteration stabilizes autonomy and competence,

[4] Table of Identified Patterns available in our dataset

Table 3. Final Structural Template for Radiant Game Design Patterns (RPs)

Template Component	Brief Description
Pattern Name	Short, conceptual name representing the essence of the solution at an abstract level.
Target Actor	Primary user of the pattern within the game development process.
Goal	Developer's main objective when applying the pattern, formulated generically across variants.
Developer Need	Structural or systemic requirements needed to support the Goal.
Player Problem	Recurring player need or difficulty addressed by the pattern.
Negative Consequences	Emotional, motivational, and behavioral impacts if the Player Problem remains unresolved.
Core Solution	Conceptual strategy explaining how the RP addresses the Player Problem and supports the Goal.
Psychological Needs Supported	SDT needs supported or kept neutral by the RP.
RP Invariants	Essential properties that must hold across all variants and levels.
Universal Application Levels	Abstract cumulative levels representing increasing satisfaction of psychological needs.
RP Variants	Contextual adaptations of the RP across genres, platforms, or systems while preserving invariants.
Expected Outcomes	Anticipated effects on player experience, well-being, engagement, and design goals.

which are frequent DP targets in games, relatedness remains a priority for future expansion, as suggested by dialogue-driven instances observed in Celeste.

God of War Ragnarök [23] featured seven instances, six for Adaptive Challenge and one of Companion. One specific instance illustrated pattern interconnectedness rather than isolated design: '*Kratos' companions (Atreus and Mimir) provide support by offering the appropriate solution after the player spends too long attempting to solve a puzzle*'. This simultaneously enacts a Companion functional variant and the Adaptive Challenge single-player variant.

In *Metroid Dread* [36], we identified eleven instances across Progression (3), Adaptive Challenge (5), and Visual Support (4). Adaptive Challenge spanned all three levels – safety, autonomy, and competence – supporting the need for a multi-level structure to capture different implementation mechanisms. *Minecraft* [37] likewise yielded eleven Creative Suite instances, reinforcing that sandbox environments embed creative freedom and experimentation as core features.

In *Shadow of the Tomb Raider* [51], nine cases reinforced the 'Safety' level of Visual Support through navigation and interaction cues. In *Teamfight Tactics*, two Companion cases were primarily aesthetic, alongside one of Progression.

Analysis detected two candidate patterns [5].: Make Your Path (secondary narratives) and Non-Playable Time (active loading/cutscene engagement). Furthermore, ten candidate variants emerged for the set of patterns. Companion was extended via Multiplayer Variants in *Teamfight Tactics* [54], supporting Relatedness through cosmetic self-expression. Adaptive Challenge revealed Competitive Balance (*TFT*) and Item Adaptation (*God of War*). Progression expanded through Fair Ranking and Environment Evolution (*Minecraft*). Visual Support incorporated Reflective Messaging (*Celeste*), Textual Incorporation, and Envi-

[5] The Candidate Patterns documentation available in our dataset

ronmental Guidance (*Tomb Raider*, *Metroid Dread*). Finally, Creative Suite in *Minecraft* generated Construction Planning and Creation Preview; these highlight how low-friction prototyping and anticipatory creation support Safety and Competence during in-game creation.

5 A Catalog of Games Radiant Patterns

The Radiant Patterns Catalog [6] is a structured knowledge system designed to bridge player well-being research and game development practice. It operationalizes the multidimensional framework of Universal Levels and Contextual Variants derived from our refinement sessions. The platform is organized around context-driven filters that allow practitioners and researchers to query patterns according to developer goals and player needs, directly reflecting Phase I findings on genre specificity and application context. The system distinguishes between Finalized RPs – those validated through formal verification – and Candidate RPs, which are undergoing co-creation. This separation enables transparency regarding pattern maturity while allowing users to choose between peer-validated standards and experimental alternatives. Filters include **variant-based** contexts (e.g., Single-Player, Multiplayer) and **development goals** (e.g., preserving progress, managing companions), all within the boundaries of the RP framework.

A core contribution of the platform is its tiered visualization of pattern levels, translating abstract psychological principles into concrete design examples. Each pattern presents its Universal Application Levels (L1, L2, L3) alongside a visual carousel of real-world game instances. These examples, combined with textual explanations, function as a semiotic bridge that illustrates how an RP manifests across levels without requiring deep SDT expertise.

To ensure scalability, the platform includes a **collaboration module** enabling repository expansion. It supports two entry types: *structured submissions*, where contributors propose new patterns or variants using the RP template vocabulary; and *problem-based contributions*, which document recurring design problems requiring ethical alternatives. All submissions are flagged as Candidates and remain visible but distinct from validated RPs until reviewed through the governance process, preserving transparency while maintaining reliability.

To safeguard integrity, we propose a governance model in which community-proposed RPs undergo a formal verification protocol modeled after the collaborative workshops described in this paper. Sessions involving both academic and industry specialists assess submissions against defined RP Invariants. Rather than binary acceptance, proposals return to contributors for iterative refinement, ensuring that candidate patterns remain accessible while being polished into finalized standards. We invite volunteers from academia and industry to participate as reviewers. Although reliance on a self-selected community may introduce selection bias or uneven SDT expertise, the documentation template serves as the primary objective check. By requiring mechanics to be mapped to specific needs, it reduces the risk of subjective gatekeeping.

[6] Radiant Patterns Catalog - radiantpatterns.vercel.app

6 Discussion: Radiant Patterns and Industry Practice

The proposed Radiant Pattern framework provides a structured foundation for both game developers and researchers to recognize positive practices that align with developer goals while also analyses their impact on player's well-being. Specifically, for game developers, the framework helps clarify the distinction between dark and radiant patterns and demonstrates how adopting radiant approaches can positively impact players, fostering more ethical and player-centered experiences while avoiding negative consequences. For game researchers, the framework offers a template to identify emerging radiant practices, analyze their impact on players, and advance researches focused on ethical game design.

By cross-referencing discussions from the collaborative workshop and refinement sessions, we were able to refine what makes a Radiant Pattern "radiant." As participants initially struggled to understand how RPs could positively affect players on the original set of patterns, the refinement became essential to explicitly show their potential impact on players' basic psychological needs and clarify which development goals they support. This refinement enables game developers to more directly adopt those pattern due to his alignment with development goals while ensuring foundational player well-being, as articulated in the operational definition of Radiant Patterns [20]. We clarify that L1 is a defensive baseline rather than an aspirational peak. It represents 'not-Dark' design, serving as the essential technical foundation upon which L2 and L3 build active support for player flourishing, using suitable mechanics in the game industry.

Unlike DPs, which are engineered for profit extraction through behavioral exploitation, RPs provide a path for sustainable player engagement grounded in need satisfaction rather than psychological coercion. We recognize that while certain DPs may generate short-term revenue gains, RPs offer a framework for sustainable player engagement. Financial success from DPs does not justify their use nor does it prevent the development of better standards. Providing alternatives such as the RPs establishes a technical basis for a more responsible industry, so short-term gains do not necessarily dictate the future of player experience. RPs provide developers with an actionable standard that respects player agency without conflicting with the inherent tensions and challenges required for engaging gameplay.

As we created new pattern titles and established universal levels of application alongside context-specific variants, we evolved the pattern set from a generalized formulations to specific applications, thereby expanding its relevance across different genres and player's profiles. This evolution was made possible by broadening the discussion beyond the initial authors [20] to a specialized panel with diverse academic backgrounds and development practices. This panel refined the logic and, yet nine participants offer a focused expert perspective rather than a global industry consensus, they enriched the process with practical examples and developer expertise. The new titles, variants, and levels produced a more actionable set of Radiant Patterns while maintaining the qualities that define them as such. Central to this evolution is the development of a standardized lexicon that bridges the communication gap between research and practice,

aiming to address the semantic fragmentation in characterizing and identifying these patterns.

The autoethnography identified RPs in released games, confirming their real-world presence. This analysis serves as a discovery method, though validating the framework's use during active development remains a future requirement. Just as DPs were identified through critical observation of manipulative practices in the game industry[56], analyzing game structures through the RPs framework allowed researchers to identify design practices that either avoid needs frustration or support the fulfillment of basic psychological needs. The structural evolution of the RP framework from generalized formulations to context-specific variants addresses a core challenge in player experience research which is the inability to account for unique, genre-specific interactions [28].

By establishing universal levels alongside these variants, the Radiant Patterns framework a contextually-specific approach, recognizing that the ethical impact of a mechanic is grounded in the unique characteristics of its gameplay environment[3]. It is important to clarify that the Universal Levels (L1-L3) are intended as a methodological baseline rather than a claim of universal experience. While the experiential reality of a patterns varies wildly between genres, such as the difference between competence in a sandbox and a soulslike, the requirement to not frustrate basic psychological needs remains the common ethical denominator. Hence, the RP variants allow the lexicon to remain stable while respecting these diverse contextual boundaries identified in the literature.

The common lexicon aims to offer a systematic alternative to rapid-response research triggered by technology panics [41]. Yet, a shared vocabulary is a tool for interaction design, not a solution for all ethical challenges. Systemic issues such as game worker welfare, harassment, and online toxicity require regulatory and cultural interventions that lie beyond the scope of interaction design [27].

Besides, identifying these new mechanics through autoethnography does not automatically qualify them as Radiant Patterns. This is why they are referred on findings as candidate RPs. Identifying an RP requires evidence of recurrence, which establishes it as a pattern[2], as well as discussion and coherence with the components defined by the framework. This distinction creates a logical need for the Radiant Patterns Catalog (Section 5). Because a single autoethnographic instance cannot confirm a pattern, and a 10-hour sample may not capture the full systemic depth of complex titles, the Catalog serves as a filter where these initial observations are flagged for further verification. The 'candidate' ideas are subjected to a verification protocol. This ensures that only those designs proven to be recurring and ethically sound are finalized, transforming subjective observations into community-validated design knowledge. Expert observation may result in false positives, reinforcing the need for the Catalog's collaborative verification protocol in which the practical application of these patterns can be observed, analyzed, and discussed collectively.

Current constraints include the absence of longitudinal testing. Establishing the RP framework and catalog provides the technical foundation needed to move toward support for positive impact. Yet, the RPs Catalog is the digital infrastruc-

ture that supports the collaborative verification process, converting community observations into systematized, positive solutions. It offers the technical basis for researchers and developers to co-create a repository of well-being-supportive design patterns. This reinforces the development of a community centered on ethical game design practices, fostering knowledge sharing and the adoption of approaches focused on player well-being.

7 Final Remarks

This work evolves the RP framework by shifting focus to the structural non-frustration of basic psychological needs through a refined lexicon, tiered levels, and standardized templates. These components establish a system to connect ethical research with game development practice. While this version lacks longitudinal data, relies on a small expert-centric sample, and excludes systemic issues like worker welfare or online toxicity, it establishes the technical foundation for future industry-led verification.

Future work focuses on testing long-term impact, expanding the relatedness dimension based on observed dialogue-driven instances, and implementing the collaborative verification protocol to transition candidate patterns into finalized standards. By providing this structure, the framework offers a technical path for the industry to move beyond identifying dark patterns toward the active co-creation of design cultures centered on player well-being

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